

Optimisation of Cosolvent Concentration for Topical Drug Delivery III – Influence of Lipophilic Vehicles on Ibuprofen Permeation

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Key Words

Ibuprofen · Mineral oil · Miglyol® 812 · Permeation · Silicone

Abstract

Previously, we have reported the effects of water, ethanol, propylene glycol and various binary and ternary mixtures of these solvents on the permeation of ibuprofen in model membranes and in skin. The present study investigates the influence of lipophilic vehicles on the transport of ibuprofen in silicone membrane and in human skin. The permeation of ibuprofen was measured from mineral oil (MO), Miglyol® 812 (MG) and binary mixtures of MO and MG. The solubility of ibuprofen was 5-fold higher in MG than in MO, however, the permeation of ibuprofen from the pure vehicles and combinations of both was comparable in silicone membrane. Additionally, there were no significant differences in skin permeation for MO and MG vehicles. When the permeation of various hydrophilic and lipophilic vehicles is considered, a trend between flux values for the model membrane and skin is evident ($r^2 = 0.71$). The findings suggest that silicone membrane may provide information on qualitative trends in skin permeation for vehicles of diverse solubility and partition characteristics.

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Introduction

The importance of excipients in determining the rate and extent of topical or transdermal delivery of actives is not widely reported in the literature. This is surprising as previous work has shown that formulation components clearly influence drug permeation [1–3]. Previously, we have investigated the influence of binary and ternary systems of hydrophilic solvents [water, ethanol and propylene glycol (PG)] on ibuprofen transport in silicone membranes and in skin [4, 5]. The influence of these solvents on the permeation of ibuprofen was studied and specific effects on diffusion and/or partition parameters were examined. For PG, enhanced permeation appears to be achieved via increased solubility and partitioning of ibuprofen into silicone or skin, while ethanol enhances ibuprofen transport chiefly by promoting drug solubility in the membranes.

The present work extends these studies to lipophilic vehicles, specifically, light mineral oil (MO) and fractionated coconut oil [Miglyol® 812 (MG)]. Light MO is a mixture of refined liquid saturated hydrocarbons obtained from petroleum which is used topically as an emollient, solvent or oleaginous vehicle [6]. As well as possessing moisture-occlusive properties [7], it has been reported as

an inert, nonpenetrating and noninteractive solvent [8, 9]. More recent studies suggest that it is absorbed into the upper layers of the skin [10]. MG is composed of mixed triglycerides of fractionated C₈–C₁₀ fatty acids derived from coconut oil. This compound has a wide variety of cosmetic and pharmaceutical applications [11, 12] but its effects on skin permeation are difficult to elucidate as it is typically employed in multicomponent preparations. To our knowledge, there is only one report in the literature concerning the effects of medium chain triglycerides on skin permeation of a model hydrophilic drug [13]. A secondary aim of the present study was to investigate any correlation between ibuprofen transport in silicone and skin for the range of solvents studied to date.

Materials and Methods

Materials

Ibuprofen was a generous gift from Wyeth Consumer Health Care (Havant, UK). HPLC grade solvents were obtained from Fisher Scientific, UK. MO was obtained from Sigma-Aldrich (UK) and MG from Sasol (Witten, Germany). The silicone membranes were obtained from Samco (Nuneaton, UK) with a measured thickness of 400 µm. All other materials were of reagent grade and were from VWR (Lutterworth, UK).

Methods

Solvent Uptake Studies

Full-thickness human skin from cosmetic surgery was obtained by informed consent and with appropriate institutional ethical approval. The epidermis was separated as reported previously [4] and stored in a freezer until required. The uptake of vehicles into silicone membrane and skin (dried for 24 h to dry weight over silica before use) was experimentally determined by a gravimetric method. The membrane was cut to an appropriate size (–1 cm² samples) and weighed using a balance (Sartorius, Epson, UK). The samples were then soaked in the vehicle for 24 h. The membranes were blotted dry with tissue paper and reweighed. Experiments were performed in triplicate, at room temperature. The amount of solvent taken up by the membrane was expressed as a weight percent. Solubility parameters for the selected vehicles were calculated using Hansen's approach [14].

Solubility Determination

Excess drug was added to each solvent or cosolvent mixture and stirred with a magnetic bar for 48 h in a water bath maintained at 32°C. Solutions were centrifuged at 10,000 rpm and analysed for drug content by HPLC or UV spectroscopy as described previously [4]. Experiments were performed in triplicate and mean values with standard deviation and coefficient of variation were calculated. Reproducibility was evaluated prior to injection of the samples and during analysis; the CV values were <5% in all experiments.

Permeation Studies in Silicone Membrane and Skin

Prior to the diffusion experiment the skin was defrosted and cut to the appropriate size. Permeation experiments were conducted with saturated solutions of ibuprofen for each vehicle using Franz diffusion cells with dimensions previously reported [4]. For all permeation experiments the receptor fluid was pH 7.4 phosphate-buffered saline and sink conditions were maintained at all time points. The diffusion experiments were performed at 32°C under occluded conditions by sealing the donor compartment with microscope cover slips. The number of replicates was at least 4.

Flux values were calculated by monitoring the cumulative amount (Q) of drug diffused as a function of time (t). The permeability coefficient (k_p) was derived from the linear portion of this profile when steady-state transport had been achieved. Partition (α = Kh) and diffusion parameters (β = D/h²) were derived by fitting the entire Q versus t profile to equation 1 [15].

$$Q = c_v \alpha \left[\beta t - \frac{1}{6} - \frac{2}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{h^2} \exp(-n^2 \pi^2 \beta t) \right] \quad (1)$$

where K is the membrane-vehicle partition coefficient of the drug, D is its diffusivity in the membrane of thickness h, α represents Kh, and β represents D/h². Differences between all values were evaluated using ANOVA with the level of significance set at p ≤ 0.05.

Results and Discussion

Solubility

Table 1 illustrates the solubility of ibuprofen in MO, MG and various binary mixtures of the solvents. The higher solubility of ibuprofen in MG relative to MO (fig. 1) reflects the similarity in their respective solubility parameters (table 2); the calculated values are 9.4 (cal/cm³)^{1/2} for MG, 7.1 (cal/cm³)^{1/2} for MO, and ibuprofen has a reported value of 9.7 (cal/cm³)^{1/2} [16]. Similar trends were observed for ibuprofen solubility in water, ethanol and PG, i.e. the closer the solubility parameter between the solute and solvent, the higher the solubility. However, with increasing amounts of MG, the solubility of ibuprofen in MG/MO systems does not increase in the exponential manner observed for ethanol/water and PG/water mixtures [4, 5].

Solvent Uptake

The uptake of selected vehicles into silicone membrane and skin is reported in table 2 and the value for water uptake is also included for comparison. The solubility parameter of silicone membrane is reported as 7.5 (cal/cm³)^{1/2} [17] and the vehicles with values closest to this are taken up to a significantly (p < 0.05) greater extent in silicone than MG and water. The solubility parameter of

Fig. 1. Relationship between flux through silicone membrane and human skin at 32°C.

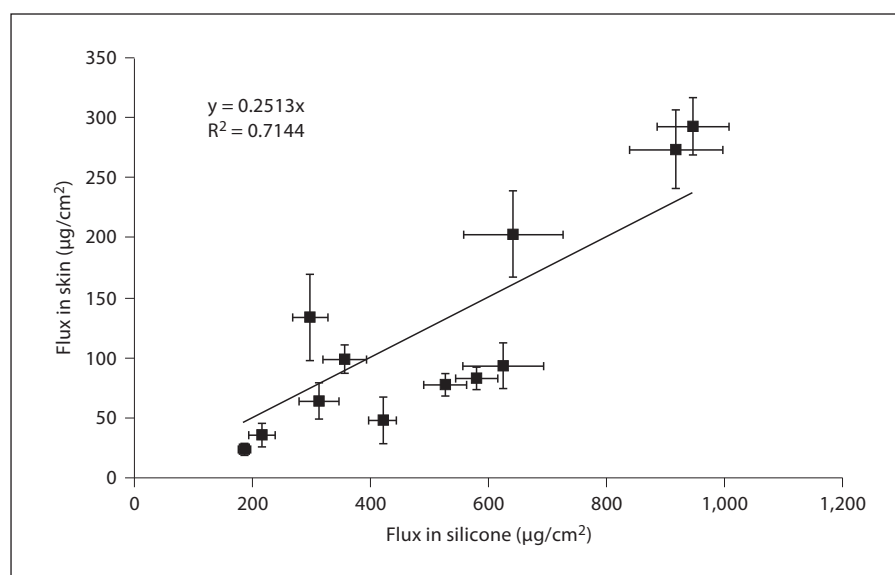


Table 1. Solubility of ibuprofen in MO and MG vehicles at 32°C (mean $\pm$ SD, n)

Mineral oil in vehicle (v/v), %	Solubility, mg/ml
0	151.60 $\pm$ 17.69
20	167.63 $\pm$ 0.43
30	125.12 $\pm$ 25.73
40	140.19 $\pm$ 33.76
60	115.73 $\pm$ 15.22
70	85.73 $\pm$ 6.26
80	53.61 $\pm$ 22.14
90	59.26 $\pm$ 7.33
100	31.99 $\pm$ 3.44

Table 2. Vehicle solubility parameter and uptake by silicone membrane and skin at 32°C (mean $\pm$ SD, n = 3)

Vehicle	Solubility parameter (cal/cm ³) ^{1/2}	Solvent uptake, % (w/w)	
		silicone	skin
MO	7.1	9.5 $\pm$ 0.6*	20.5 $\pm$ 5.9
MO/MG (50/50)	8.3	9.6 $\pm$ 1.1*	14.1 $\pm$ 9.8
MG	9.4	2.3 $\pm$ 0.3	21.9 $\pm$ 4.4
Water	23.4	1.6 $\pm$ 0.8	89.8 $\pm$ 22.6

* p $\leq$ 0.05: significantly different from MG and water.

the skin is thought to be close to 10 (cal/cm³)^{1/2} [18]. While there are no significant differences in the extent to which MO, MO/MG and MG are taken up by skin, it is interesting to note that the weight of skin is almost doubled after water uptake. The extent of water uptake is also significantly greater than for the other solvents into skin. The homogeneous nature of silicone when compared to skin a priori dictates that it will be a more sensitive discriminator of solvent/membrane interaction.

Permeation Studies in Silicone Membrane

The permeation of ibuprofen through silicone membrane, using MO and MG vehicles of varying composition, is reported in table 3. The highest permeation rates (>600 μ g/cm²/h) were observed for 60/40, 70/30 MO/MG, pure MO and pure MG formulations. The flux values for these formulations were significantly higher than for water (185.5 μ g/cm²/h), pure PG (297.1 μ g/cm²/h), 60/40 PG/water (355.4 μ g/cm²/h) and 70/30 PG/water systems (443.3 μ g/cm²/h) which were previously reported [5]. Partition and diffusion parameters did not show any distinct trends (data not shown). The corresponding k_p values (table 3) do not change significantly, with the exception being the pure MO vehicle. The results suggest that it is the uptake of MO which is dictating the more favourable flux values. For MG, it is likely that the enhanced solubility of ibuprofen in the vehicle compensates for the lower uptake of the MG, resulting in flux values comparable with the MO vehicle.

Permeation Studies in Skin

There were no significant differences between the permeation rates of ibuprofen through human skin for pure MO, 50:50 MO/MG and pure MG (table 4). The flux values are much lower than for the corresponding systems through silicone membrane, as observed for permeation of other formulations of ibuprofen in silicone and skin [4, 5]. The skin permeation of ibuprofen in these lipophilic vehicles is comparable to that from PG or PG/water formulations [5] but lower than that from ethanol-rich formulations [4]. As for the silicone data, the k_p value for MO is significantly greater than that observed for the MG vehicle. There were no significant differences for the respective vehicle uptake into skin (table 2), suggesting that the increased solubility (MG > MO) of the ibuprofen in the solvent taken up by the stratum corneum largely explains the permeation enhancement.

Silicone and Skin Permeation

Considering the findings for all single, binary and ternary hydrophilic and lipophilic vehicles reported to date [4, 5] and excluding the pure ethanol system, linear regression of silicone and skin permeation data (fig. 1) gives a correlation coefficient of 0.71. Correlations between transport in silicone membranes and skin for simple aqueous solutions or solutions with small proportions of ethanol have previously been reported [19, 20]. More complex systems were investigated by Iervolino et al. [21], who analyzed the relationship between silicone and skin permeation for a range of saturated and supersaturated solutions of ibuprofen in PG/water vehicles. A correlation coefficient of 0.96 was noted. These authors also noted that the flux in silicone was approximately 2.3 times that in skin. Figure 1 indicates that the flux in silicone is approximately 4 times that in human skin. However, it is important to note that the vehicles investigated to date [4, 5] encompass solvents with a much wider spectrum of physicochemical properties than those previously investigated for silicone – skin permeation. The relationship observed suggests that silicone membranes may be used to probe the activities of solvents and solvent mixtures with considerable diversity in solubility and partition characteristics. It is important to note that silicone is not proposed as an alternative to other models of the stratum corneum which contain structured lipids [22]. For the lipophilic vehicles, drug permeation through solvent-enriched silicone membranes appears to mirror drug permeation through solvent-enriched skin epidermal membranes. Silicone membranes are far less complex than other theoretical models of human skin.

Table 3. Steady-state flux values and permeability coefficients for the permeation of ibuprofen from saturated formulations containing MO and/or MG through silicone membrane (mean $\pm$ SD, n = 4)

MO/MG (v/v)	k_p , cm/h $\times 10^2$	J, $\mu\text{g}/\text{cm}^2/\text{h}$
0/100	0.41 ± 0.05	623.8 ± 67.7
10/90	0.39 ± 0.15	439.9 ± 23.9
20/80	0.34 ± 0.14	569.6 ± 63.5
30/70	0.45 ± 0.08	573.5 ± 55.0
40/60	0.29 ± 0.11	409.4 ± 13.6
50/50	–	311.9 ± 33.8
60/40	0.58 ± 0.06	675.1 ± 66.8
70/30	0.75 ± 0.07	646.9 ± 9.9
80/20	0.97 ± 0.21	519.1 ± 20.2
90/10	0.77 ± 0.31	457.0 ± 53.5
100/0	$1.81 \pm 0.28^*$	579.4 ± 35.3

* $p < 0.05$: significantly different from all other vehicles.

Table 4. Steady-state flux values and permeability coefficients for the permeation of ibuprofen from saturated formulations containing MO and MG 812 through human epidermis (mean $\pm$ SD, n = 5)

MO/MG (v/v)	k_p , cm/h $\times 10^2$	J, $\mu\text{g}/\text{cm}^2/\text{h}$
0/100	0.06 ± 0.01	93.6 ± 19.1
50/50	–	64.5 ± 15.2
100/0	$0.26 \pm 0.08^*$	83.1 ± 9.2

* $p < 0.05$: significant difference.

Conclusions

Saturated solutions of ibuprofen in the lipophilic vehicles studied resulted in relatively high flux values in silicone compared to PG vehicles, but these values are considerably lower than those observed for ethanol and ethanol-rich vehicles. In skin, ibuprofen permeation from MO, MG or a mixture of both was also comparable to ibuprofen skin permeation from PG or mixed PG/water vehicles but lower when compared to ethanol/water formulations, where the proportion of ethanol was $\geq 50\%$. The ability of MG to enhance drug solubility likely explains the permeation enhancement in silicone, whereas the greater MO uptake is responsible for similar ibuprofen permeation values. In skin, the similar permeation values are in line with the solvent uptake for both vehicles, suggesting that permeation enhancement is dictated

by the vehicle uptake in skin. A linear correlation is obtained from analysis of the model membrane and skin permeation data for the range of hydrophilic and lipophilic vehicles studied to date. The findings confirm the utility of silicone membrane in evaluating the influence of vehicle activity on drug permeation as a precursor to conducting such studies in skin.

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References

- Wotton PK, Mollgaard B, Hadgraft J, Hoelgaard A: Vehicle effect on topical drug delivery. III. Effect of azone on the cutaneous permeation of metronidazole and propylene glycol. *Int J Pharm* 1985;24:19–26.
- Trottet L, Merly C, Mirza M, Hadgraft J, Davis AF: Effect of finite doses of propylene glycol on enhancement of in vitro percutaneous permeation of loperamide hydrochloride. *Int J Pharm* 2004;274:213–219.
- Otto A, Wiechers JW, Kelly CL, Hadgraft J, du Plessis J: Effect of penetration modifiers on the dermal and transdermal delivery of drugs and cosmetic active ingredients. *Skin Pharmacol Physiol* 2008;21:326–334.
- Watkinson RM, Herkenne C, Guy RH, Hadgraft J, Oliveira G, Lane ME: Influence of ethanol on the solubility, ionisation and permeation characteristics of ibuprofen in silicone and human skin. *Skin Pharmacol Physiol* 2009;22:15–21.
- Watkinson RM, Guy RH, Hadgraft J, Lane ME: Optimisation of cosolvent concentration for topical drug delivery. II. Influence of propylene glycol on ibuprofen permeation. *Skin Pharmacol Physiol* 2009;22:225–230.
- Abramowitz R, O'Donoghue D: Mineral oil, light; in Rowe RC, Sheskey PJ, Weller PJ (eds): *Handbook of Pharmaceutical Excipients*, ed 4. Washington, Pharmaceutical Press and American Pharmaceutical Association, 2003, pp 398–399.
- Tranner F, Berube G: Mineral oil and petrolatum – reliable moisturizers. *Cosmet Toilettries* 1978;93:81.
- Roberts MS, Anderson RA: The percutaneous absorption of phenolic compounds: the effect of vehicle on the penetration of phenol. *J Pharm Pharmacol* 1975;27:599–605.
- Evans JR, Hughes C: In vitro methodology for investigating quantitative relationships between skin permeation and compound structure. *J Pharm Pharmacol* 1986;38:68P.
- Backhouse L, Dias M, Gorce JP, Hadgraft J, McDonald PJ, Wiechers JW: GARField magnetic resonance profiling of the ingress of model skin-care product ingredients into human skin in vitro. *J Pharm Sci* 2004;93:2274–2283.
- Iannuccelli V, Coppi G, Sergi G, Mezzena M, Scalia S: In vivo and in vitro skin permeation of butyl methoxydibenzoylmethane from lipospheres. *Skin Pharmacol Physiol* 2008;21:30–38.
- Pouton CW, Porter CJH: Formulation of lipid-based delivery systems for oral administration: materials, methods and strategies. *Adv Drug Deliv Rev* 2008;60:625–637.
- Cornwell PA, Tubek J, van Gompel HAHP, Little CJ, Wiechers JW: Glyceryl monocaprylate/caprate as a moderate skin penetration enhancer. *Int J Pharm* 1998;171:243–251.
- Hansen CM: *Hansen Solubility Parameters. A User's Handbook*. Boca Raton, CRC Press, 2000.
- Diez-Sales O, Watkinson AC, Herraiz-Dominguez M, Javaloyes C, Hadgraft J: A mechanistic investigation of the in vitro human skin permeation enhancing effect of Azone®. *Int J Pharm* 1996;129:33–40.
- Bustamante P, Pena MA, Barra J: The modified extended Hansen method to determine partial solubility parameters of drugs containing a single hydrogen bonding group and their sodium derivatives: benzoic acid/Na and ibuprofen/Na. *Int J Pharm* 2000;194:117–124.
- Most CF: Co-permeant enhancement of drug transmission rates through silicone rubber. *J Biomed Mater Res* 1972;6:3–14.
- Liron Z, Cohen S: Percutaneous absorption of alkanolic acids. I. A study of operational conditions. *J Pharm Sci* 1984;73:534–537.
- Ginoz S, Rey S, Boss G, Bunge AL, Guy RH, Carrupt P-A, Reist M, Testa B: Quantitative structure-permeation relationships for solute transport across silicone membranes. *Pharm Res* 2002;19:1622–1629.
- Wasdo SC, Juntunen J, Devarajan H, Sloan KB: A correlation of flux through a silicone membrane with flux through hairless mouse skin and human skin in vitro. *Int J Pharm* 2009;373:62–67.
- Iervolino M, Cappello B, Raghavan SL, Hadgraft J: Penetration enhancement of ibuprofen from supersaturated solutions through human skin. *Int J Pharm* 2001;212:131–141.
- Kessner D, Ruettinger A, Kiselev MA, Wartewig S, Neubert RHH: Properties of ceramides and their impact on the stratum corneum structure: a review. Part 2: stratum corneum lipid model systems. *Skin Pharmacol Physiol* 2008;21:58–74.

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